Differentiation by first principles

1 Consider the function $f\left(x\right)=x^{3}$.

a Complete the limiting chord process for $f\left(x\right)=x^{3}$ at the point $x=3$ by filling in the table of values. Round your answers to four decimal places.

a	b	$h=b-a$	$\dfrac{f\left(b\right)-f\left(a\right)}{b-a}$
3	4	1	
3	3.5		
3	3.1		
3	3.05		
3	3.01		
3	3.001		
3	3.0001		

b Use the limiting chord process to calculate the instantaneous rate of change for the remaining values of x in the table:

x	1	2	3	4	5
Instantaneous rate of change $f(x)$ at x	3	12		48	

c Find the instantaneous rate of change of $f\left(x\right)$ at any point x.

2 Consider the function $f\left(x\right)=x^{2}$.

a Complete the limiting chord process for $f\left(x\right)=x^{2}$ at the point $x=3$ by filling in the table of values:

a	b	$h=b-a$	$\dfrac{f\left(b\right)-f\left(a\right)}{b-a}$
3	4	1	
3	3.5		
3	3.1		
3	3.05		
3	3.01		
3	3.001		
3	3.0001		

b Use the limiting chord process to calcu[redacted]he instantaneous rate of change for the remaining values of x in the table:

x	1	2	3	4	5
Instantaneous rate of change $f(x)$ at x	2	4		8	

c Find the instantaneous rate of change of $f\,(x)$ at any point x.

3 Consider the function $f\,(x) = 2x^2$.

a Complete the limiting chord process for $f\,(x) = 2x^2$ at the point $x = 3$ by filling in the table of values. Round your answers to four decimal places.

a	b	$h = b - a$	$\dfrac{f\,(b) - f\,(a)}{b - a}$
3	4	1	
3	3.5		
3	3.1		
3	3.05		
3	3.01		
3	3.001		
3	3.0001		

b Use the limiting chord process to calculate the instantaneous rate of change for the remaining values of x in the table:

x	1	2	3	4	5
Instantaneous rate of change of $f(x)$ at x	4	8		16	

c Hence, find the instantaneous rate of change of $f\,(x)$ for any given value of x.

4 For each of the following functions:

i Find $f(x+h)$.

ii Find $f(x+h)-f(x)$.

iii Find $\dfrac{f(x+h)-f(x)}{h}$.

a $f(x)=3x+2$

b $f(x)=8-x$

c $f(x)=2x-3$

d $f(x)=-5x+4$

e $f(x)=5-x^2$

f $f(x)=x^2+3x+6$

g $f(x)=\dfrac{1}{x}$

h $f(x)=\dfrac{1}{7x}$

i $f(x)=3x^2-4x+2$

5 For each of the following functions:

i Find $f(x+h)$ in expanded form.

ii Find $f(x+h)-f(x)$.

iii Find $\dfrac{f(x+h)-f(x)}{h}$.

iv Hence, find $f'(x)$ by evaluating $\lim_{h\to 0}\left(\dfrac{f(x+h)-f(x)}{h}\right)$.

a $f(x)=x^2$

b $f(x)=4x^2$

c $f(x)=-6x^2$

6 Find the derivative of each of the following functions from first principles:

a $f(x)=6$

b $f(x)=-6$

c $f(x)=5x$

d $f(x)=-2x$

e $f(x)=2x+3$

f $f(x)=x^2-3$

g $f(x)=4x^2+5$

h $f(x)=x^2-3x$

i $f(x)=2x^2-3x$

j $f(x)=4x^2-3x-5$

k $f(x)=\sqrt{x}$

l $f(x)=\sqrt{x-5}$

Gradients

7 Consider the function $f(x) = 4x$.

- **a** Is the gradient of the function constant or variable?
- **b** Find the gradient of the function using the points $(1, 4)$ and $(7, 28)$ that lie on the function.
- **c** Find the gradient of the function from first principles.

8 Consider the function $f(x) = (x+6)(x+2)$.

- **a** Find the gradient function of $f(x) = (x+6)(x+2)$ from first principles.
- **b** Find the gradient at the point where $x = 3$.

9 Consider the function $y = \sqrt{x}$.

- **a** Find the gradient function of $f(x) = \sqrt{x}$ from first principles.
- **b** Find the gradient at the point where $x = 25$.

10 Consider the function $f(x) = 7 - x^2$.

- **a** Find the gradient function $f'(x)$ using first principles.
- **b** Hence, calculate the gradient of $f(x)$ at the point where $x = -1$.

11 Consider the function $f(x) = x^2 + 3x + 4$.

- **a** Find the gradient function $f'(x)$ using first principles.
- **b** Hence, calculate the gradient of $f(x)$ at the point where $x = 4$.

12 Consider the function $f(x) = x^3$.

- **a** Find the gradient function $f'(x)$ using first principles.
- **b** Hence, calculate the gradient of $f(x)$ at the point where $x = 2$.

13 Consider the function $f(x) = x^4$.

- **a** Find the gradient function $f'(x)$ using first principles.
- **b** Hence, calculate the gradient of $f(x)$ at the point where $x = -5$.

14 Find the gradient of $f(x) = 3x^2$ at the poi[redacted]48) from first principles.

15 Find the gradient of $f(x) = 3x^3$ at the point $(-4, -192)$ from first principles.

16 Find the gradient of $f(x) = 2x(x-6)$ at the point $(2, -16)$ from first principles.

17 Find the gradient of $f(x) = 3x(x-2)^2$ at the point $(4, 48)$ from first principles.

18 Consider the function $y = \dfrac{1}{x}$.

a Find the gradient function of $y = \dfrac{1}{x}$ from first principles.

b Find the x-values of the two points on the function at which the gradient is $-\dfrac{1}{16}$.

19 Consider the function $f(x) = \dfrac{1}{2x-4}$.

a Find $f'(x)$ from first principles.

b Hence, find $f'(4)$.